## Unit: IV
## File Organizations

Storing the files in certain order is called file organization. The main objective of file organization is:

- Optimal selection of records i.e. records should be accessed as fast as possible.
- Any insert, update or delete transaction on records should be easy, quick and should not harm other records.
- No duplicate records should be induced as a result of insert, update or delete.
- Records should be stored efficiently so that cost of storage is minimal.

**Some of the file organizations are:**

- **Sequential File Organization**
- **Indexed Sequential Access Method**
- **Hash/ Direct File organization**
- **Multilist File Organization**
- **Inverted File organization**
- **B+ Tree File Organization**

**Sequential Access File Organization:**

- Storing and sorting in contiguous block within files on tape or disk is called as sequential access file organization.
- In sequential access file organization, all records are stored in a sequential order. The records are arranged in the ascending or descending order of a key field.
- Sequential file search starts from the beginning of the file and the records can be added at the end of the file.
- In sequential file, it is not possible to add a record in the middle of the file without rewriting the file.
- **Advantages:**
    - It is simple to program and easy to design.
    - Files can be easily stored in magnetic tapes i.e. cheaper storage mechanism.
    - Simple design
- **Disadvantages:**

  - Time wastage as we cannot jump on a particular record that is required, but we have to move in a sequential manner which takes our time.
  - Sorting file method is inefficient as it takes time and space for sorting records.

**Hash/ Direct Access File Organization:**

- Direct access file is also known as random access or relative file organization.
- In this method of file organization, hash function is used to calculate the address of the block to store the records.
- The hash function can be any simple or complex mathematical function.
- In this, each record is stored randomly irrespective of the order they come. Hence, this method is also known as Direct or Random file organization.
- **Advantages:**
  - Records need not be sorted after any of the transaction. Hence, the effort of sorting is reduced in this method.
  - Since, block address is known by hash function, accessing any record is very faster. Similarly updating or deleting a record is also very quick.
  - This method can handle multiple transactions as each record is independent of other i.e. since there is no dependency on storage location for each record, multiple records can be accessed at the same time.
  - It is suitable for online transaction systems like online banking, ticket booking system etc.

**Indexed Sequential Access File Organization:**

- Indexed sequential access file combines both sequential file and direct access file organization.
- In indexed sequential access file, records are physically stored on the disk in groups. Within each group the records are stored sequentially by primary key.
- Records in an ISAM file may be indexed in two different ways:
  - The entire file is read/searched sequentially using the primary key; and

    - the record is accessed via the index using the key value provided in each index entry, the address is located within the index and the record is retrieved.
- ISAM file combines some of the good features of indexed and sequential files. The ISAM file requires relatively smaller disk storage space, as it maintains records in the sequence on each track of the file.
- The major disadvantage of the index-sequential organization is that as the file grows, performance deteriorates rapidly because of overflows and consequently there arises the need for periodic reorganization. Reorganization is an expensive process and the file becomes unavailable during reorganization.

**B+ Tree File Organization:**

- It uses a tree like structure to store records in File.
- It uses the concept of key indexing where the primary key is used to sort the records.
- For each primary key, an index value is generated and mapped with the records.
- An index of a record is the address of record in the file.
- B+ Tree is very much similar to binary search tree, with the only difference that instead of just two children, it can have more than two children.
- All the information is stored in leaf node and the intermediate nodes acts as pointer to the leaf nodes.
- The information in leaf nodes always remain a sorted sequential linked list.
- **Pros:**
    - Tree traversal is easier and faster.
    - Searching becomes easy as all records are stored only in leaf nodes and are sorted sequential linked list.
    - There is no restriction on B+ tree size. It may grow/shrink as the size of data increases/decreases.
- **Cons:**
    - Inefficient for static tables.

**Multi list File Organization:**

- The basic approach to providing the linkage between an index and the file of data records is called multi list organization.
- A multi list file maintains an index for each secondary key.

- The index of secondary key contains, instead of a list of primary keys related to that secondary key, only one primary key value related to that secondary key. The record will be linked to other records containing the same secondary key in the data file.

**Inverted File Organization:**

- In inverted file organization, a linkage is provided between an index and the file of data records.
- In the inverted file organization, one index is maintained for each key attribute of the record. The index file contains the value of the key attribute followed by the addresses of all the records in the main file with the same value of the key attribute.
- The inverted file organization requires three kinds of files to be maintained: the main file, the directory files and the index files.
- The directory file contains the value of the key attributes and the pointer to the first record in the index file where the addresses of all the records in the main file with the value of the key attribute are contained. There is a directory file for each key attribute.
- Inverted file is very useful where the list of records with specified values of key attributes is required.